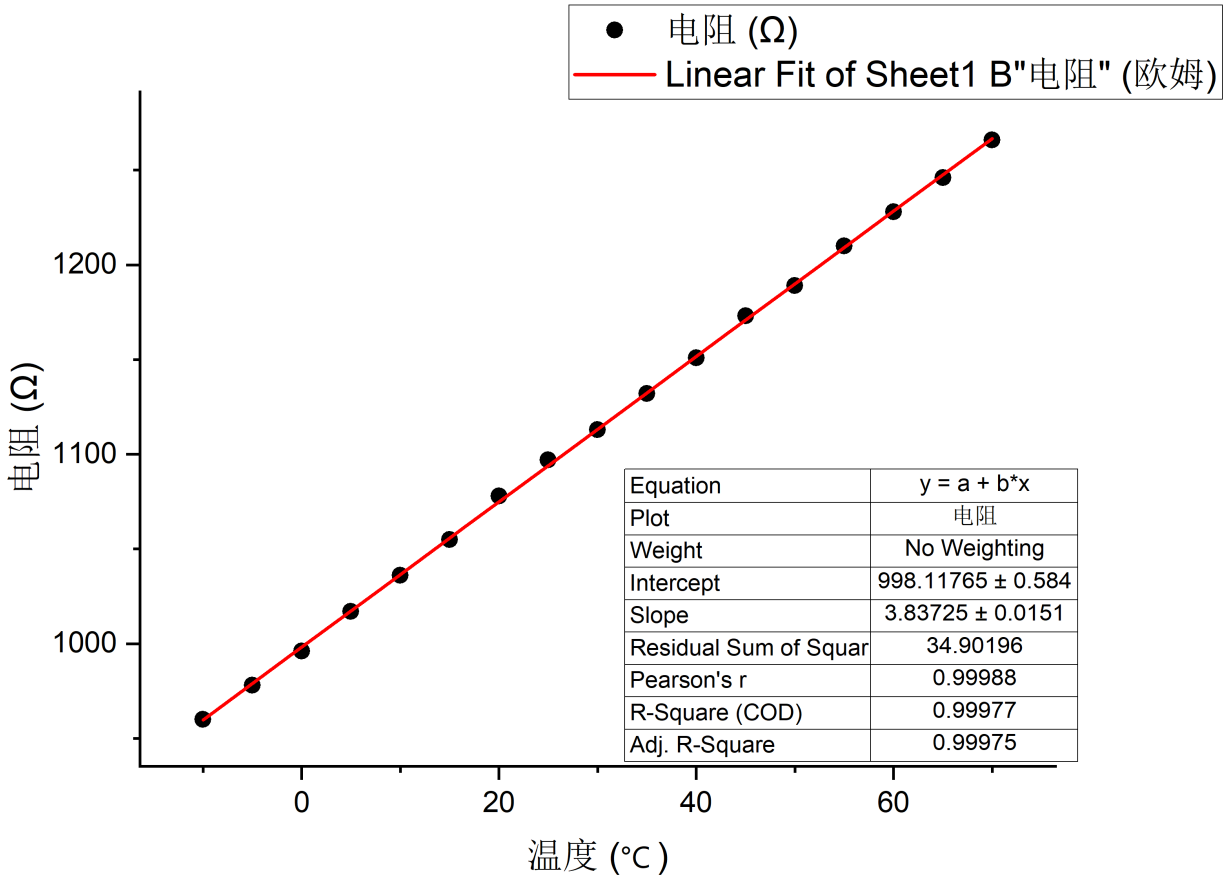
图一：RTD的 $R-T$ 曲线



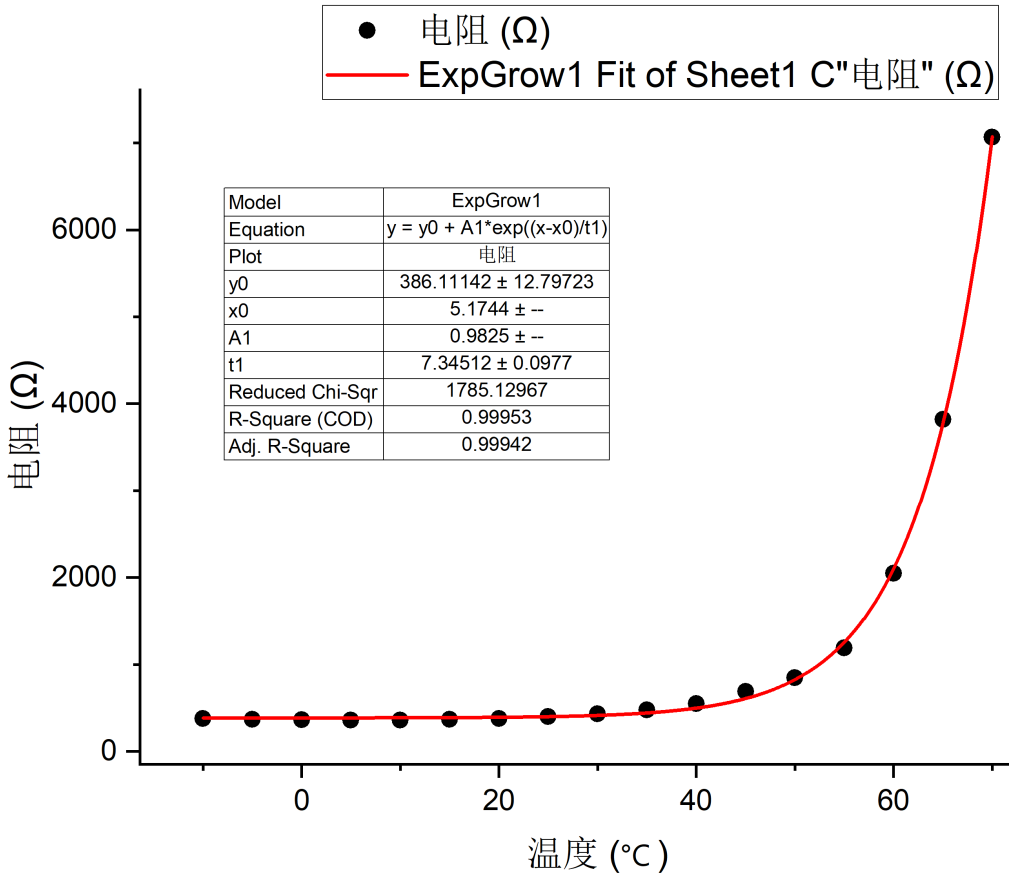
由软件拟合的结果可知，其标定公式为

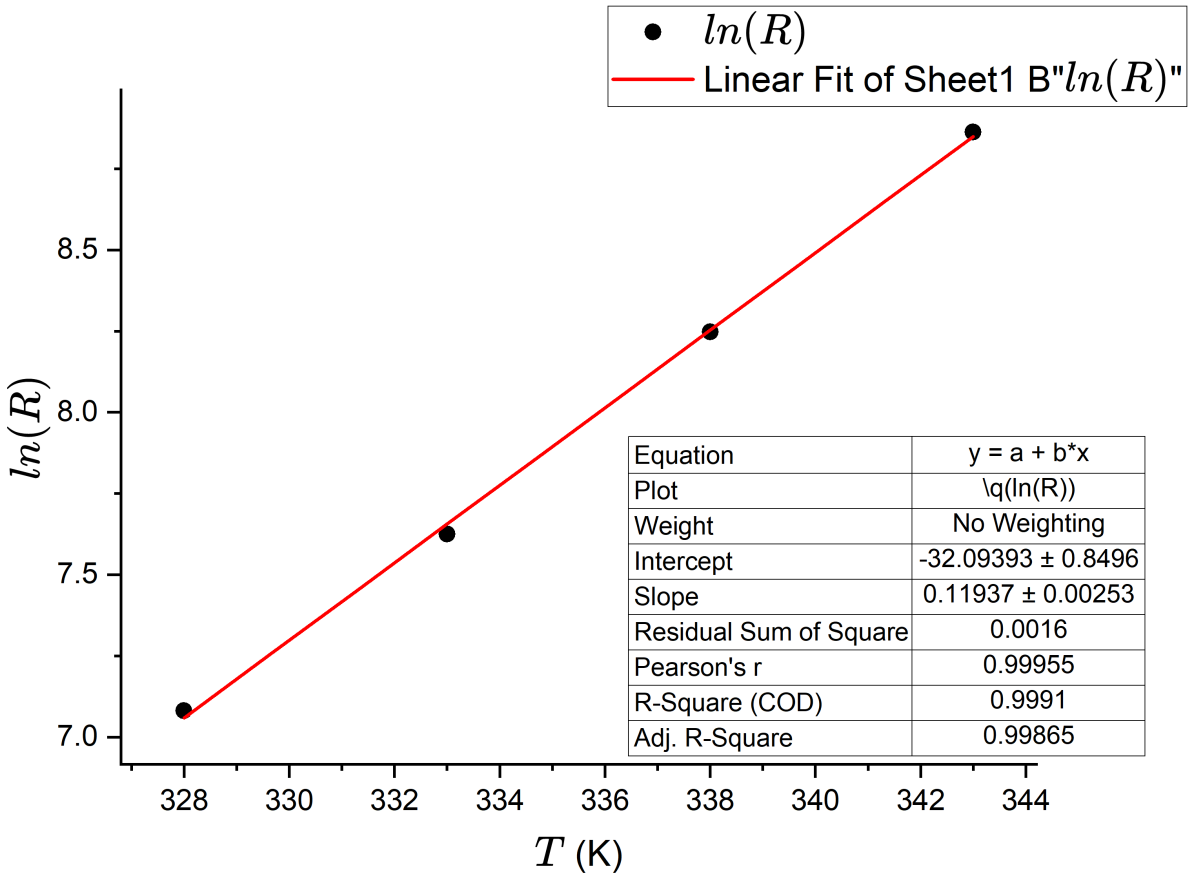
$$R=(3.837\pm0.015)T+(998.12\pm0.58)$$

与 R_0 的参考值 1000Ω 相比，相对误差为

$$\delta=\frac{|1000-998.12|}{1000}\times100\%=0.188\%$$

图二：PTC的 $R-T$ 曲线和 $\ln R-T$ 曲线

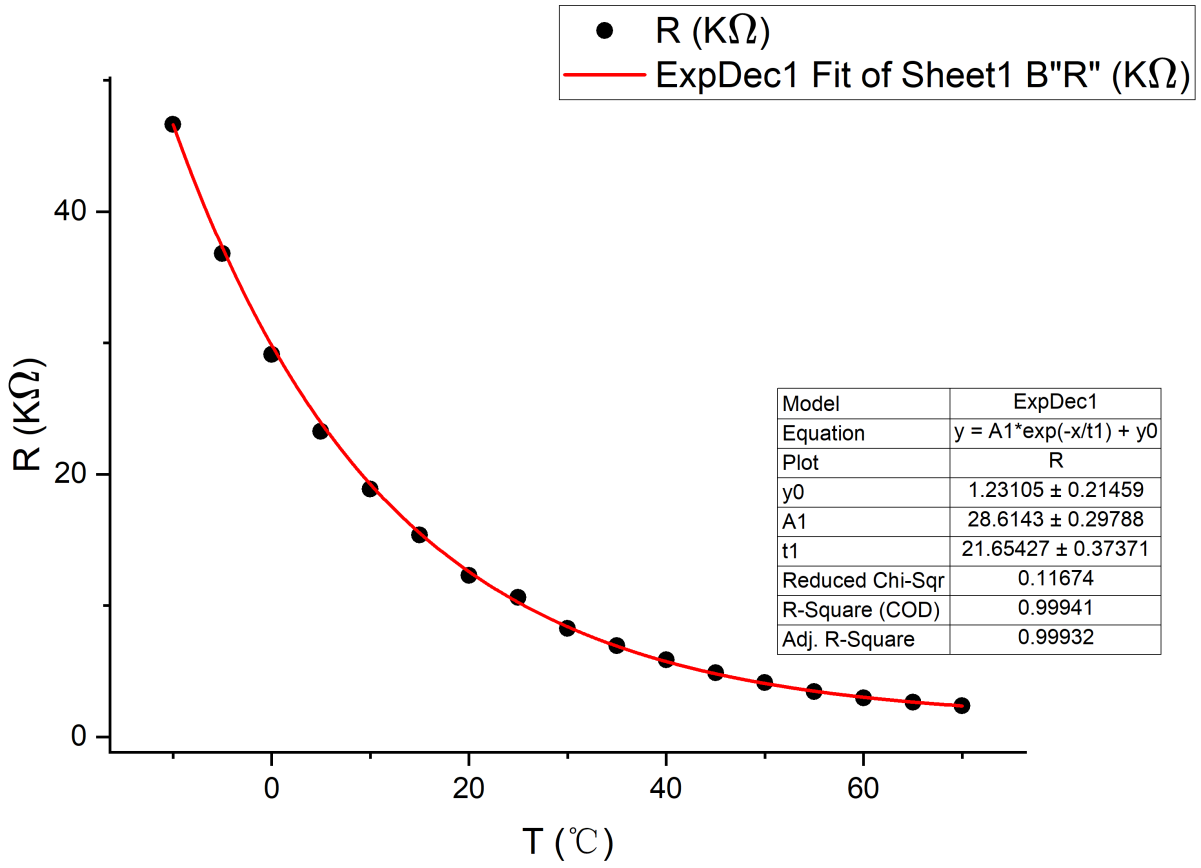

电阻 (Ω)
ExpGrow1 Fit of Sheet1 C"电阻" (Ω)
Model ExpGrow1
Equation y = y0 + A1*exp((x-x0)/t1)
Plot 电阻
y0 386.11142 ± 12.79723
x0 5.1744 ± --
A1 0.9825 ± --
t1 7.34512 ± 0.0977
Reduced Chi-Sqr 1785.12967
R-Square (COD) 0.99953
Adj. R-Square 0.99942
电阻 (Ω)
温度 (°C)
0
2000
4000
6000
0
20
40
60


ln(R)
Linear Fit of Sheet1 B"ln(R)"
Equation y = a + b*x
Plot \q(ln(R))
Weight No Weighting
Intercept -32.09393 ± 0.8496
Slope 0.11937 ± 0.00253
Residual Sum of Square 0.0016
Pearson's r 0.99955
R-Square (COD) 0.9991
Adj. R-Square 0.99865
ln(R)
T (K)
7.0
7.5
8.0
8.5
328
330
332
334
336
338
340
342
344

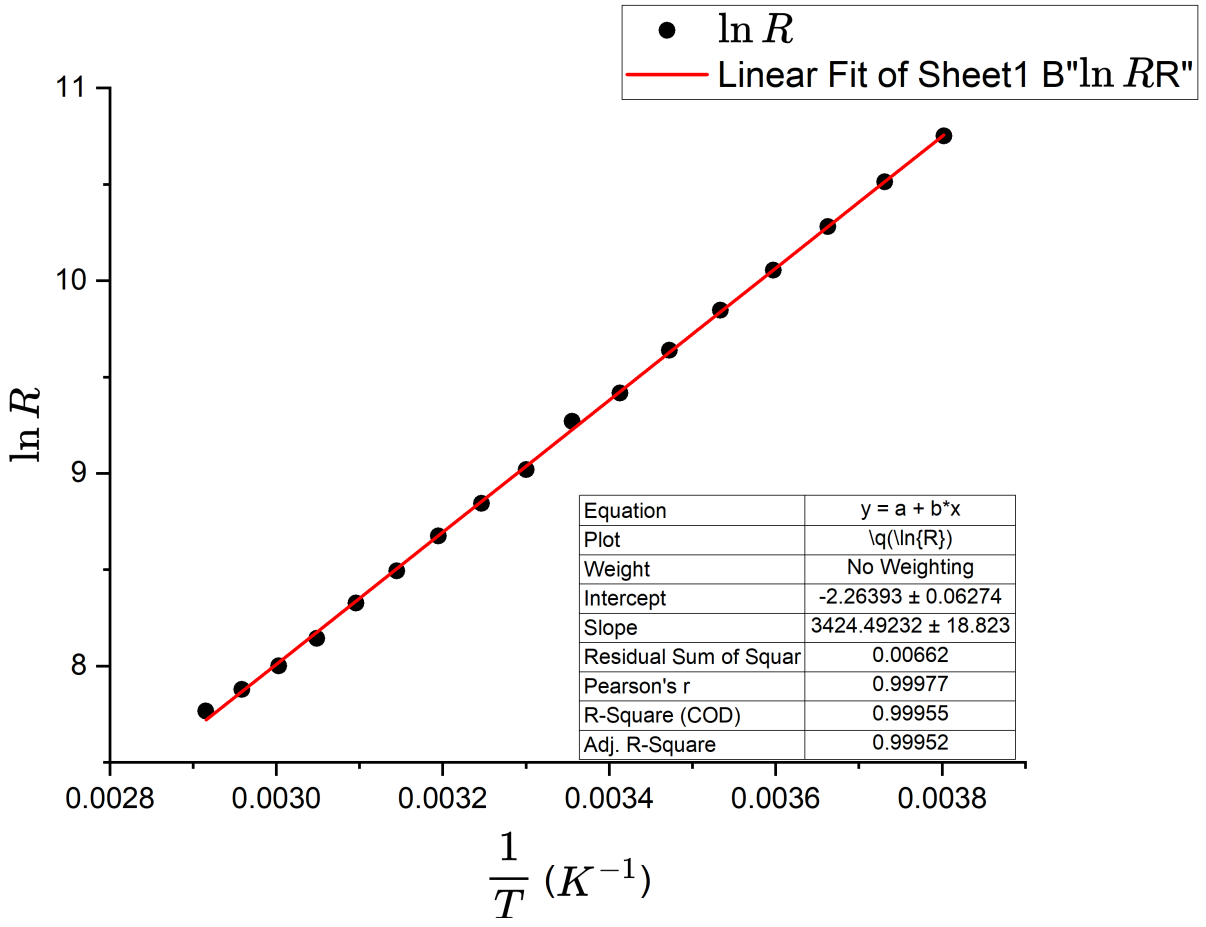
选取 55~70℃ 的区间进行拟合，得到的直线斜率为 PTC 的材料常数

$$\beta_P = (0.1193 \pm 0.0025)K$$

图三： NTC的 $R-T$ 曲线和 $\ln R - \frac{1}{T}$ 曲线



Model	ExpDec1
Equation	y = A1*exp(-x/t1) + y0
Plot	R
y0	1.23105 ± 0.21459
A1	28.6143 ± 0.29788
t1	21.65427 ± 0.37371
Reduced Chi-Sqr	0.11674
R-Square (COD)	0.99941
Adj. R-Square	0.99932



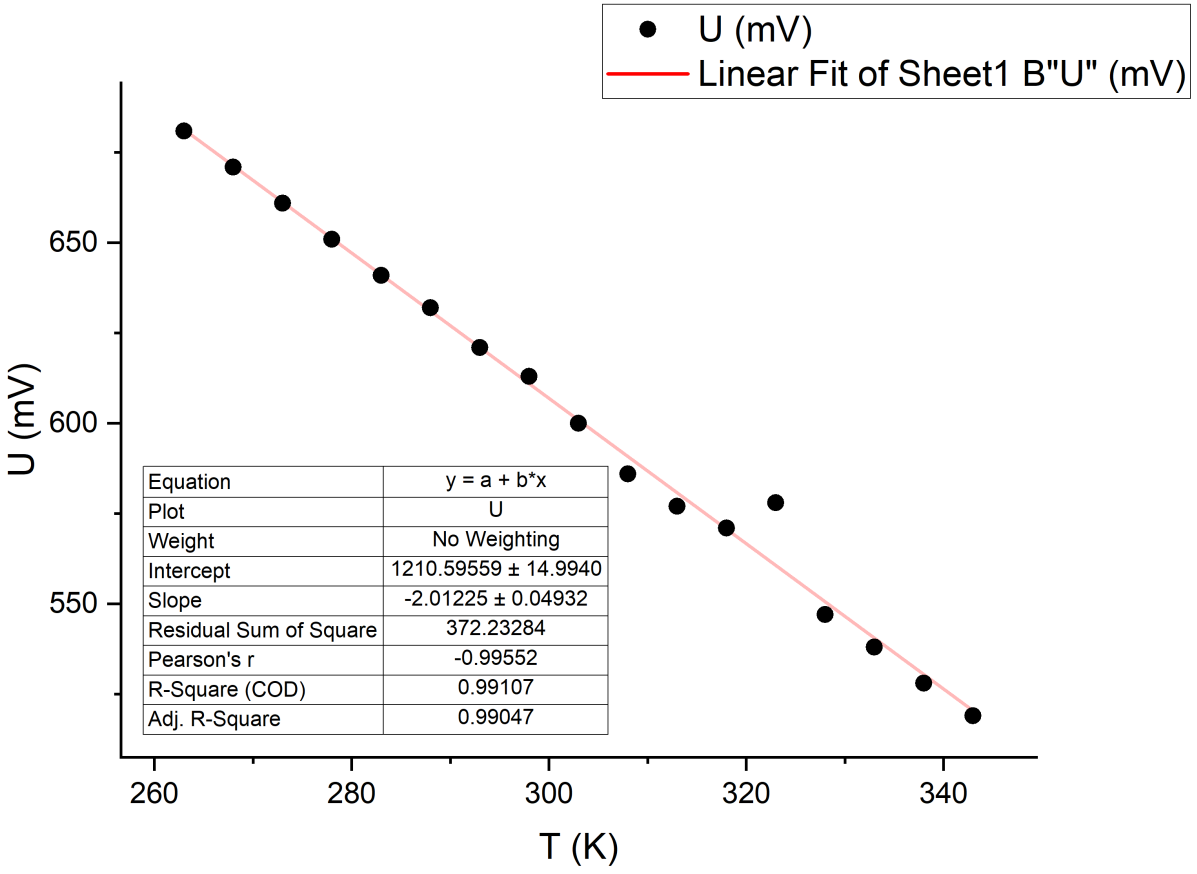
由软件拟合的结果可知，其标定公式为

$$\ln R = (3424.49 \pm 19)\frac{1}{T} - (2.264 \pm 0.063)$$

材料常数等于 $\ln R - \frac{1}{T}$ 曲线的斜率，所以

$$\beta_n = (3424.49 \pm 19)K$$

图四：PN结的 $U-T$ 曲线



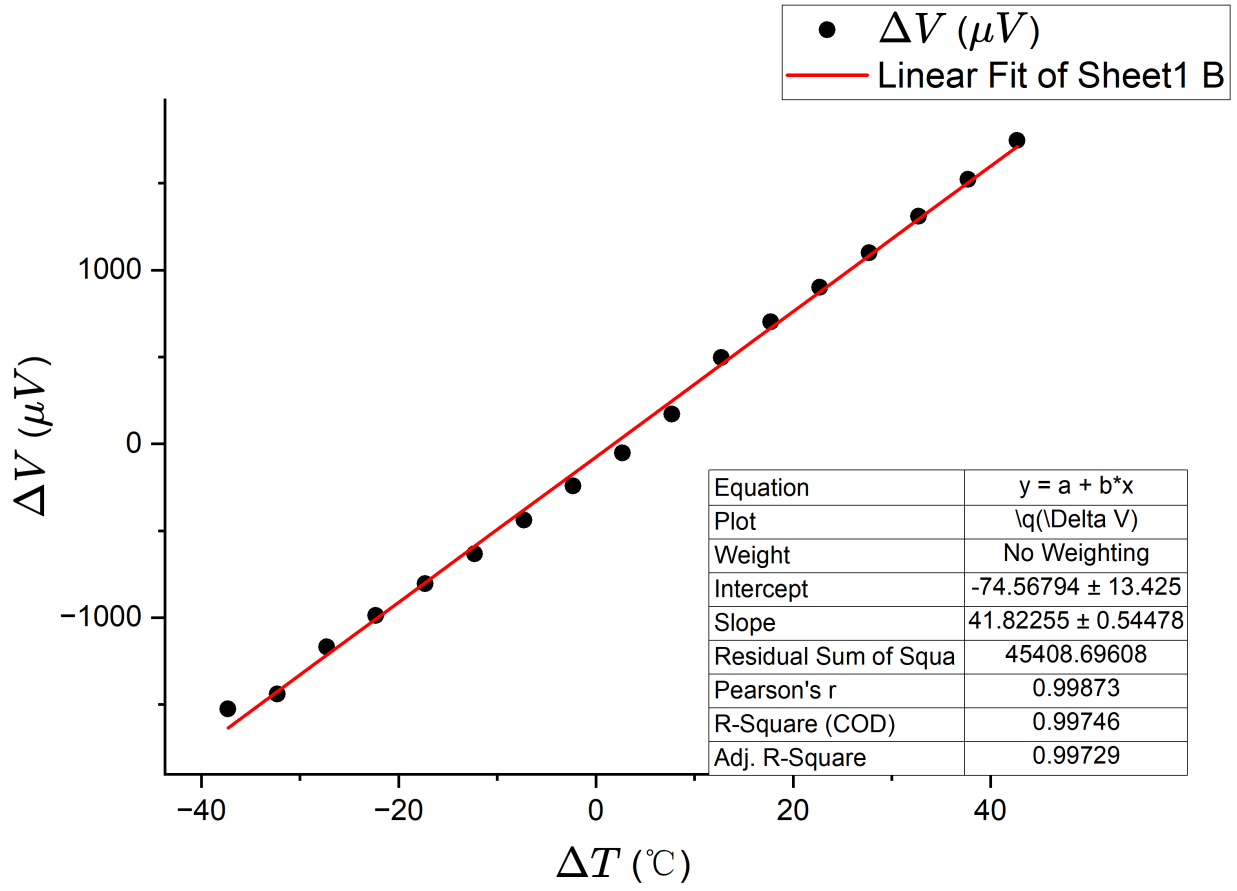
由软件拟合结果可知，PN结的电压温度系数为直线的斜率，即

$$K=(-2.012\pm0.049)mV/K$$

相对误差为

$$\delta=\frac{|(-2.012)-(-2)|}{|-2|}\times100\%=0.6\%$$

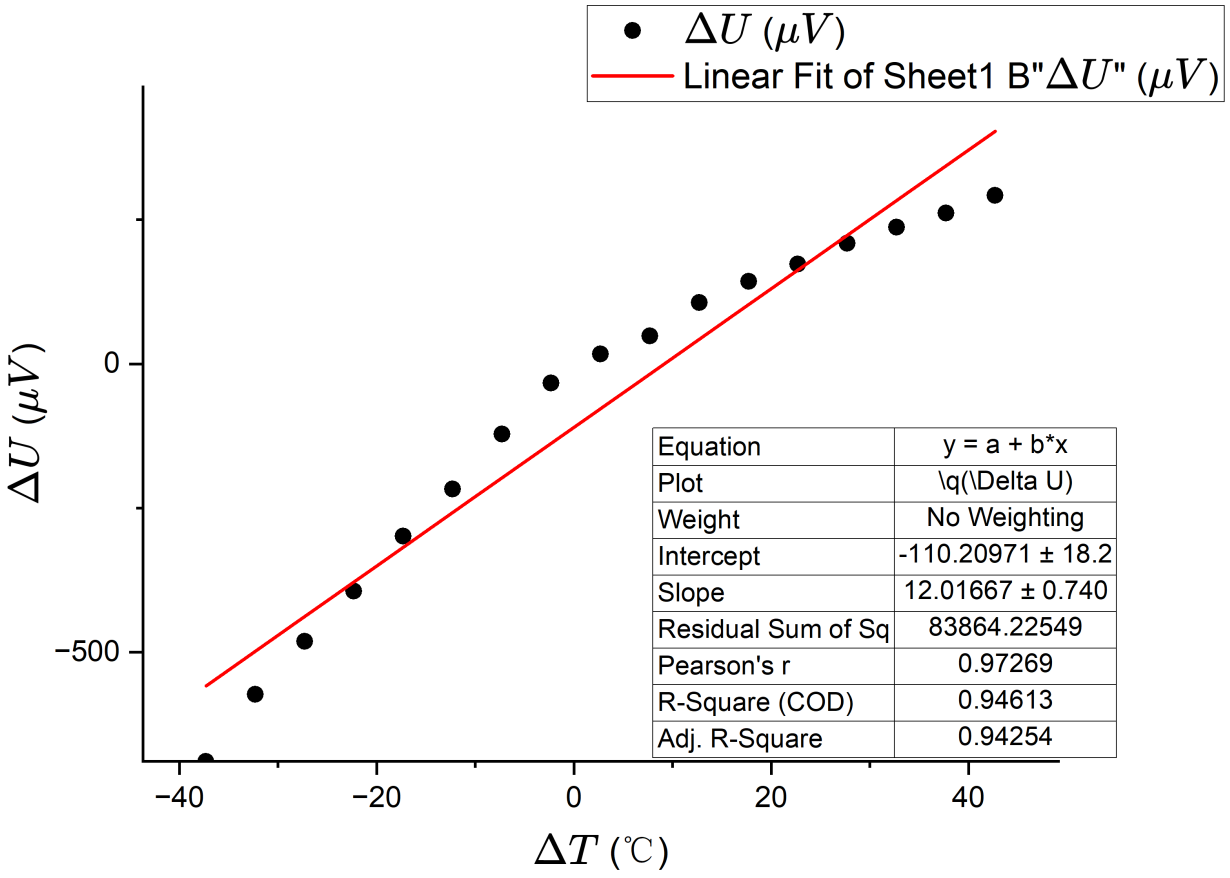
图五：热电偶温度传感器的 ΔU与ΔT 曲线



由软件拟合结果可知，拟合曲线的斜率即为电压温度系数

$$S = (41.82 \pm 0.54)\mu V/℃$$

图六： 热电堆型红外辐射温度传感器的 ΔU与ΔT 曲线



由软件拟合结果可知，拟合曲线的斜率即为电压温度系数

$$S = (12.02 \pm 0.74)\mu V/℃$$